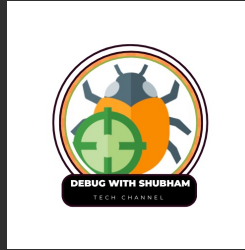


# *Debug With Shubham Notes*



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# NQT 2025



## TCS Interview for ECE/EEE Students

### Digital Role 2025

Do They Know You're Non-CS?

Tell me about yourself (customized for ECE)

Why IT after ECE/EEE?

Basic Python / Coding logic (reverse string, factorial, prime)

OOPs Concepts (Class, Object, Inheritance)

Project Explanation

Optional: Basic DBMS/Networking questions

HR Questions – Why TCS? Are you ready to relocate?

# What You Can Skip (Don't Waste Time)

Graphs, Trees, Dynamic Programming

OS memory management

Deep Networking protocols

Competitive coding  
(Unless you've already studied them!)

## What to Focus On Instead

20 Python logic questions(vo question mai hi dunga apko)

OOPs with Python examples

Prepare 1 project well

Learn to answer “Why IT after ECE?”

Smart Tip for Answering “Why IT after ECE?”

“Sir, during my graduation I developed interest in coding through Python. I like logical problem-solving and I'm confident that my ECE background combined with IT skills makes me a strong learner in this space.”

## **Prep Timeline (If you have 1 week):**

Day 1–2: Python basics + logic

Day 3: OOP + project prep

Day 4–5: HR + mock interviews

Day 6–7: Resume polish + mock test

# 20 Python Coding Q&A

## 1. Reverse a String

Question: Write a Python program to reverse a string

```
s = "hello"
print(s[::-1]) # Output: 'olleh'.
```

## 2. Factorial of a Number

Question: Find the factorial of a number using a loop.

```
n = 5
fact = 1
for i in range(1, n + 1):
    fact *= i
print(fact) # Output: 120
```

## 3. Check Prime Number

Question: Check if a number is prime or not.

```
num = 17
is_prime = True
if num <= 1:
    is_prime = False
else:
    for i in range(2, int(num**0.5) + 1):
        if num % i == 0:
            is_prime = False
            break
print("Prime" if is_prime else "Not Prime")
```

## 4. Palindrome Check (String or Number)

Question: Check if the input is a palindrome.

```
s = "madam"
if s == s[::-1]:
    print("Palindrome")
else:
    print("Not Palindrome")
```

## 5. Fibonacci Series

Question: Print Fibonacci series up to N terms.

```
n = 5
a, b = 0, 1
for _ in range(n):
    print(a, end=" ")
    a, b = b, a + b
```

## 6. Count Characters in String

```
s = "aabbcc"
from collections import Counter
print(Counter(s))
```

## 7. Find Largest in List

```
arr = [3, 7, 2, 9]
print(max(arr)) # Output: 9
```

### 8. Sum of Digits

```
num = 123
s = sum(int(d) for d in str(num))
print(s) # Output: 6
```

### 9. Armstrong Number

```
num = 153
order = len(str(num))
sum_ = sum(int(d)**order for d in str(num))
print("Armstrong" if sum_ == num else "Not Armstrong")
```

### 10. Even/Odd Count in List

```
arr = [1, 2, 3, 4, 5]
even = len([x for x in arr if x % 2 == 0])
odd = len(arr) - even
print(f"Even: {even}, Odd: {odd}")
```

### 11. Count Vowels in a String

```
s = "hello world"
vowels = "aeiouAEIOU"
count = sum(1 for c in s if c in vowels)
print(count)
```

### 12. Check Leap Year

```
year = 2024
is_leap = (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0)
print("Leap Year" if is_leap else "Not Leap Year")
```

### 13. Swap Two Numbers

```
a, b = 5, 10
a, b = b, a
print(a, b) # Output: 10 5
```

### 14. Sum of N Natural Numbers

```
n = 10
print(n * (n + 1) // 2) # Output: 55
```

### 15. Find Duplicates in List

```
lst = [1, 2, 3, 2, 4, 1]
seen = set()
dups = set(x for x in lst if x in seen or seen.add(x))
print(dups)
```

### 16. Remove Punctuation from String

```
import string
s = "Hello, world!"
cleaned = "".join(c for c in s if c not in string.punctuation)
print(cleaned) # Output: Hello world
```



### 17. Print Table of a Number

`n = 5`

```
for i in range(1, 11):  
    print(f"{n} x {i} = {n*i}")
```

### 18. Reverse an Integer

`num = 1234`

`rev = int(str(num)[::-1])`

`print(rev)`

### 19. Check Even or Odd

`n = 4`

`print("Even" if n % 2 == 0 else "Odd")`

### 20. Calculate Power (Exponentiation)

`base, exp = 2, 3`

`print(base ** exp) # Output: 8`

# OOPs with Python examples

## 1. Class and Object

A class is a blueprint; an object is an instance of that class.

```
class Person:
    def __init__(self, name):
        self.name = name

    def greet(self):
        print("Hello", self.name)

p = Person("Alice")
p.greet() # Output: Hello Alice
```

## 2. Inheritance

Child class inherits properties and methods of the parent class

```
class Animal:
    def speak(self):
        print("Animal speaks")

class Dog(Animal):
    def speak(self):
        print("Dog barks")
```

```
d = Dog()
d.speak() # Output: Dog barks
```

### 3. Encapsulation

Hiding internal details using private variables (with \_\_).

```
class BankAccount:
    def __init__(self):
        self.__balance = 1000 # private

    def get_balance(self):
        return self.__balance

b = BankAccount()
print(b.get_balance()) # Output: 1000
```

### 4. Polymorphism

Same function behaves differently based on object.

```
class Cat:
    def sound(self):
        print("Meow")

class Dog:
    def sound(self):
        print("Bark")

for animal in (Cat(), Dog()):
    animal.sound()
```

## 5. Abstraction

Hiding complex logic and showing only essentials. (Use base class with NotImplementedError or abc module.)  
from abc import ABC, abstractmethod

```
class Shape(ABC):  
    @abstractmethod  
    def area(self):  
        pass
```

```
class Circle(Shape):  
    def __init__(self, radius):  
        self.radius = radius  
  
    def area(self):  
        return 3.14 * self.radius * self.radius
```

```
c = Circle(5)  
print(c.area()) # Output: 78.5
```

Do TCS ask SQL or DBMS questions to Non-CS students (ECE/EEE)?

Yes, but only if...

You have mentioned SQL/DBMS in your resume

You've done a project involving a database

The interviewer feels you're confident in basic coding and wants to test further

🧠 So what's the rule for Non-CS students?

| Topic  | Asked When                              | Should You Prepare?     |
|--------|-----------------------------------------|-------------------------|
| *SQL*  | Mentioned in resume or project          | ✓ Learn basics          |
| *DBMS* | Only if the interviewer dives deeper    | ✓ Theory only           |
| *DSA*  | Not expected unless you claim expertise | ✗ Can skip advanced DSA |

## Minimum SQL Knowledge (if preparing):

SELECT, WHERE, GROUP BY, ORDER BY, COUNT(\*)

Simple JOIN between two tables

Finding top salary, duplicate values, etc.

## Minimum DBMS Theory:

What is DBMS?

Difference between DBMS & RDBMS

What is Primary Key / Foreign Key

What is Normalization?

## What if I didn't study SQL or DBMS?

### You can honestly say:

“Sir, I haven't worked much with databases so far, but I am open to learning and picking it up quickly on the job.”

No negative marking if you're honest + confident.

### Final Advice:

If not on your resume — you can safely skip deep SQL/DBMS

If you have time — learn just the basics to feel confident

### Conclusion & Motivation:

"Don't panic if you're from ECE — many students from non-CS backgrounds get selected every year. Be honest, show learning attitude, and you'll crack it!"